

Module full title	Computer Science Final Project
SITS Module Code	6COSC023W
Credit level	Level 6
Length	Year Long
UK credit value	40
ECTS credit value	20
College	Design, Creative and Digital Industries
School	Computer Science and Engineering
Host programme of study	BSCSS02F (BSc Computer Science FT)
Progression and assessment board	COMENG - Computing UG PAB
Pre-requisites	None
Co-requisites	None
Study abroad	No
Special features	Franchised to IIT Sri Lanka - 6COSC023C
Access restrictions	N/A
Are the module learning outcomes delivered, assessed or supported through an arrangement with organisation(s) other than the University of Westminster:	No
Summary of module content	The module allows students to demonstrate in applied, written and oral forms the effective use of competences required to achieve the degree of study. It requires students to combine previously acquired knowledge and techniques, with new knowledge/ideas gained from investigation and research, and produce an extended piece of work related to their degree of study. It involves the conceptualisation, design, implementation and evaluation of a substantial piece of software, process, model or experimental study.

Module aims

The aims of the module are to provide the student with an understanding of the skills, methodology and processes to enable the student to implement a project starting from a 'natural language specification' through to successful project implementation, documentation, and deployment. On completion of the module, the student will be equipped to work autonomously on real-world challenges.

Assessment Methods

Rank	Assessment type	Assessment name	Weighting	Minimum mark required	Qualifying set (where the minimum mark required applies across multiple assessments)
	Presentation	Project Proposal and Requirements Specification	15		
	Presentation	Interim Progress Demonstration	15		
	Project	Final Year Project	70	30	

Synoptic assessment

This module contains a synoptic learning assessment (report and viva) that combines elements of student learning from different parts of the course in order to demonstrate and assess the accumulated knowledge and understanding gained in the execution of an autonomous project.

Learning outcomes

By the end of the module the successful student will be able to:

- LO1 Select, justify and apply appropriate methods, techniques and tools for tackling a large problem;
- LO2 Develop a project plan, organise own activities and manage time and workload within the given timescale;
- LO3 Collect and analyse project requirements using appropriate tools and techniques;
- LO4 Research and collate relevant information from various sources and critically evaluate the findings taking any commercial and economic context into account;
- LO5 Work autonomously to learn new skills/enhance existing knowledge and produce deliverables agreed with the supervisor;
- LO6 Identify and take into account any EDI, legal, social, ethical and professional issues related to the problem and its solution including sustainability considerations;
- LO7 Produce an extended piece of practical work (software, process, model, experimental study);
- LO8 Produce a coherent report that documents and critically evaluates the practical work, the new skills and the effectiveness of the project plan;
- LO9 Demonstrate and defend their work at a viva voce examination.

Course outcomes the module contributes to

BSc Computer Science: L6.3, L6.4, L6.5, L6.6
 BEng Software Engineering: L6.1, L6.4, L6.5

Indicative syllabus content

- Developing a project idea.
- Background and literature review.
- Project planning and management.
- Project Lifecycle – choosing a suitable lifecycle or methodology for the project.
- Ethical, legal and sustainability considerations.
- Project requirements – capturing and modelling project requirements, qualities and constraints using appropriate research methods.
- Design - experimental design, application design, user interface, data visualisation, prototyping.
- Experimentation and implementation.
- Testing.
- Report writing, evidencing, documenting, critical evaluation, presentation and style.

Teaching and learning methods

The T&L rationale of the FYP module is to support students develop into independent learners, apply the knowledge and skills gained from the degree they study, gain new knowledge and address the challenges imposed by the FYP. The students are allocated a supervisor that guides and directs them throughout the duration of the module. The supervisor allocation is carried out on the basis of matching students' preference and supervisors' expertise, interests and availability. Based on their interest the students select a project topic. Students are also exposed to project topics proposed by academics, reflecting on-going research at the School of Computer Science and Engineering and providing the opportunity for student/academic/industry collaborations. The project is agreed between the student and the supervisor to ensure it complies with the FYP requirements. Students work independently, they form, execute and adapt a project plan depending on developing circumstances, in consultation with their supervisor.

Teaching and learning methods include independent learning, using material and guidance provided on Blackboard, scheduled workshops and regular supervisory meetings as depicted in the following table and independent study.

Workshops related to all aspects of the FYP lifecycle are planned to offer hands-on contact time support and feedback to students. Students are expected to attend regular meetings with their supervisor to review progress and discuss various aspects of the project. Actions planned during the meetings are recorded in a project management journal.

Activity type	Category	Student learning and teaching hours *
Lecture	Scheduled	
Seminar	Scheduled	
Tutorial	Scheduled	
Project supervisor	Scheduled	12
Demonstration	Scheduled	
Practical classes and workshops	Scheduled	
Supervised time in studio/workshop	Scheduled	
Fieldwork	Scheduled	
External visits	Scheduled	
Work based learning	Scheduled	
Scheduled online learning	Scheduled	18
Other learning	Scheduled	
Total scheduled		30
Placement	Placement	
Independent study	Independent	370
Total student learning and teaching hours		400

* hours per activity type are indicative and subject to change

Assessment rationale: Why has this assessment been used for this module?

The assessment covers both the product delivered by the student at the end of the module, and the process by which this product was developed. The module is assessed by:

- **Project Proposal and Requirements Specification** - presentation (up to 20 minutes) of a project proposal that enables students to engage with a project idea. The proposal will require the student to carry out background research, aims and objectives, and produce a detailed statement of requirements (LO2, LO3, LO4).
- **Interim Progress Demonstration** - a presentation (up to 20 minutes) on the initial output from design and practical work (e.g. demonstration of a software prototype, development tools, research progress) that enables students to progress toward the development of their Final Year Project deliverables. (LO1, LO2, LO3, LO4, LO5).
- **Final Year Project** – a report (up to 12,000 words excluding diagrams and/or code), final output from practical work and viva that enable students to demonstrate the competence they gained in delivering a self-initiated project, communicate effectively in written and oral forms and the competences required to achieve the degree of study. (LO1-LO9)

Project Proposal and Requirements Specification (PPRS) (15%): a presentation that includes aims, objectives and scope of the project; definition of the problem; initial literature review; requirements gathering process; a detailed list of requirements; time schedule; bibliography.

Interim Progress Demonstration (IPD) (15%): a presentation that includes stakeholders; modelling requirements and relevant diagrams; listing the project requirements; proposed design of the software/system to be developed, or plan of study and tools to be used in case an experimental study or theoretical work is being conducted. The **IPD** also includes a practical demonstration of output – a prototype of the software/system to be developed capturing key functionality. This shall highlight and discuss the student's progress towards the project's objectives against an original plan. The student will be given written feedback on the report and the prototype/plan of study from their supervisor.

Final Year Project (FYP) (70%): comprises of a written report; practical work and viva. A successful final project is a combination of practical work and a report detailing the development and a critical analysis of the resulting research/software/application. The FYP describes the work carried out and it is accompanied by the software/application/research results. The FYP is assessed for logical structure, coherence and relevance. It is also assessed for clear documentation on the scope, depth and complexity of the work; the background research; the development of new and existing skills; the selection and application of appropriate techniques; the different aspects of the development process; any legal, social, ethical, professional and sustainability issues relevant to the work; the reflection and critical evaluation of the work. The practical output is assessed in terms of level of complexity, stability, robustness, performance and originality.

Students are expected to attend a 1-hour viva. Two members of staff, normally the project supervisor and another member of staff, assess the student's performance. During the viva, the student is expected to demonstrate their practical output (i.e. software, working prototype, research outcomes) and to be tested on their understanding of relevant issues. Additionally, students are expected to demonstrate understanding of the wider context and propose further work. Formative feedback is provided throughout the project lifecycle to support students to progress and refine the development of their project. All formative assessments are designed to help students progressing toward the delivery of the next (time wise) Summative Assessment and ultimately the delivery of the Final Year Project.

Students are required to regularly demonstrate the extent of the development of their work. The project supervisors provide regular feedback to assist the on-going development of the project.

Assessment criteria: What criteria will be used to assess my work on this module?

University Grade Descriptors are a benchmark point of reference, they are contextualised using specific subject specialist criteria specific to a particular assessment.

[University Grade Descriptors](#)

Criteria for assessments are designed with reference to the University's generic criteria to measure students' ability to meet the learning outcomes of a module. Specifically, within this module you will find detailed grading descriptors as part of each assessment.

To succeed students must demonstrate competence in autonomously planning and producing a substantial practical piece of work (software, process, model, experimental study), present it in a well-constructed and coherent report that documents and critically evaluates the work carried out, and defend it in a viva.

Deliverables include a completed and functioning software application and a report documenting the sound application of software methodology, design, implementation. The quality of the project work will be assessed based on the following criteria:

- Background research and understanding of the project area.
- The development of existing and new skills.
- The identification of relevant legal, social, ethical, professional and sustainability issues.
- The level of independence and self-management ability.
- The complexity and the execution of practical work.
- Academic writing skills.
- Oral presentation skills.

To achieve high marks, students must demonstrate a greater degree of expertise and knowledge, and in particular the ability to develop an ambitious or novel application/working prototype. The students must use appropriate techniques and demonstrate furthering the development of existing skills and the acquisition of new skills. The students must also demonstrate a greater degree of academic writing skills, reflection and critical evaluation abilities.

Project Assessment Threshold Criteria

Grade	Indicative meaning
90-100	LO1 – LO9. The project is complete, with major critical additions by student. An exemplary piece of work carried out to an extremely high professional standard. Work shows an extremely deep understanding of the problem and evidence of highly creative and original thought is evident throughout. Publishable project. LO1. Exemplary work is shown in demonstrating the selection, justification and the application of appropriate methods, techniques and tools in tackling problems relating to the project. LO2. Exemplary work displayed in project management developing a project plan that scheduled the student's own activities and time relating to the project activities and timescale. LO3. Exemplary work demonstrated in the elicitation of the requirements of the project. Exceptionally clear analysis and specification of the problem being solved demonstrated. LO4. Exceptionally coherently presented background research interleaved with student's own creative commentaries and constructive criticisms of broader issues and related literature to the project. Student is able to present very challenging and complex background material in a precise and logical, yet concise manner. LO5. Very high technical difficulty and a very challenging project. Exceptional development of existing skills. Exceptional development of new skills. LO6. The student has demonstrated an exceptional understanding and wider consideration to legal, social, ethical, professional and sustainability issues that relate to the project. LO7. Exceptionally well presented design and high-level structure of the software produced. Comprehensive and well-grounded justification of the main design decisions. Exhaustive testing of the software supported with personal comments for the requirements of the specific tests. LO8. Exceptionally well-written and well-organised professional report. All main themes and issues are clearly identified and presented in a professional way. Exceptionally clear appraisal of the project, including clear well-structured conclusions. LO9. The student demonstrated a very deep understanding of both the theory and the related practical material in the viva. The software produced is exceptional, with novel and original features. The software is stable and robust and has excellent performance.

80-89	LO1 – LO9. The project is complete, with major critical additions by student and shows that s/he can work on this subject quite independently in a research environment. An outstanding piece of work carried out to a high professional standard. LO1. Outstanding work is shown in demonstrating the selection, justification and the application of appropriate methods, techniques and tools in tackling problems relating to the project. LO2. Outstanding work displayed in project management developing a project plan that scheduled the student's own activities and time relating to the project activities and timescale. LO3. Outstanding work demonstrated in the elicitation of the requirements of the project. A very clear analysis and specification of the problem being solved demonstrated. LO4. Outstanding coherently presented background research. Student is able to demonstrate an outstanding critical analysis of broader issues and related literature to the project. LO5. A technically difficult and a challenging project. Excellent development of existing skills. Excellent development of new skills. LO6. The student has demonstrated an excellent understanding and wider consideration to legal, social, ethical, professional and sustainability issues that relate to the project. LO7. Well presented design and high-level structure of the software produced. Comprehensive and well-grounded justification of the main design decisions. Exhaustive testing of the software. LO8. An excellent well-written and well-organised report. All main themes and issues are clearly identified and presented. Exceptionally clear appraisal of the project, including clear well-structured conclusions.
70-79	LO1 – LO9. The project is complete, with major critical additions by student. An excellent piece of work carried out to a very high standard. LO1. Excellent work is shown in demonstrating the selection, justification and the application of appropriate methods, techniques and tools in tackling problems relating to the project. LO2. Excellent work displayed in project management developing a project plan that scheduled the student's own activities and time relating to the project activities and timescale. LO3. Excellent work demonstrated in the elicitation of the requirements of the project. A very clear analysis and specification of the problem being solved demonstrated. LO4. Excellently coherently presented background research. Student is able to demonstrate an outstanding critical analysis broader issues and related literature to the project. LO5. High technical difficulty and a challenging project. Significant development of existing skills. Significant development of new skills. LO6. The student has demonstrated an excellent understanding and wider consideration to legal, social, ethical, professional and sustainability issues that relate to the project. LO7. Clearly presented design and high-level structure of the software produced. Well- grounded justification of the main design decisions. Exhaustive testing of the software. The software has all the required and expected features. The software is stable and robust and has excellent performance. LO8. A well-written and well-organised report. Main themes and issues are clearly identified. Very clear appraisal of the project, including clear well- structured conclusions. LO9. The student is competent in technical matters and understands the theory as demonstrated in the viva. The student is able to solve own problems independently.

60-69	LO1 – LO9. The project is complete, with some critical additions by student. A very good piece of work carried out to a very good standard. LO1. Very good work is shown in demonstrating the selection, justification and the application of appropriate methods, techniques and tools in tackling problems relating to the project. LO2. Very good work displayed in project management developing a project plan that scheduled the student's own activities and time relating to the project activities and timescale. LO3. Very good work demonstrated in the elicitation of the requirements of the project. A good clear analysis and specification of the problem being solved demonstrated. Well-written analysis and specification of the problem being solved. However, the specification is not always sufficiently detailed. LO4. Very good coherently presented background research. Student is able to demonstrate an outstanding critical analysis broader issues and related literature to the project. Background research present, although at times the presentation lacks coherence. Some lacking in breadth of literature review. LO5. A moderately technically difficult and a challenging project. Good development of existing skills. Good development of new skills. Has needed some help by supervisor. LO6. The student has demonstrated a good understanding and wider consideration to legal, social, ethical, professional and sustainability issues that relate to the project. LO7. The high-level structure of the software produced and its design are clearly presented. Main design decisions are described, but some justifications are not convincing. Testing of the software almost complete. LO8. Well-written and well-organised report. Main themes and issues are clearly identified. Clear appraisal of the project, although the conclusions are not clearly structured. LO9. The student demonstrated an understanding of the practical material in the viva but not of the all the theoretical aspects. The software has almost all the required and expected features. The software has good performance and is mostly stable and robust.
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50-59	LO1 – LO9. The project is complete, with some work above 'textbook' standard. A good piece of work carried out to a good standard. LO1. Good work is shown in demonstrating the selection, justification and the application of appropriate methods, techniques and tools in tackling problems relating to the project. LO2. Good work displayed in project management developing a project plan that scheduled the student's own activities and time relating to the project activities and timescale. LO3. Good work demonstrated in the elicitation of the requirements of the project. A clear analysis and specification of the problem being solved demonstrated in most parts. Acceptable analysis and specification of the problem being solved. Both the analysis and specification are not sufficiently well described, but it is still possible to have a relatively clear picture of what has been solved. LO4. Background research presently well. Most of the relevant background material is adequately presented. At times the presentation lacks coherence. However, the student is able to demonstrate a good critical analysis of broader issues and related literature to the project. LO5. A technically sound and a fairly challenging project. Good development of existing skills. Some development of new skills. Has needed some help by supervisor. LO6. The student has demonstrated a fair understanding and wider consideration to legal, social, ethical, professional and sustainability issues that relate to the project. LO7. The high-level structure of the software produced and its design are adequately presented. However, they are either too brief. Main design decisions are described, but the justifications are not convincing. Almost adequate testing of the software but with a number of missing minor tests. LO8. Adequately written report. Most of the main themes and issues are clearly identified. Most appraisal issues are addressed but the conclusions are brief and at times not informative. Evidence of critical analysis in a few places. □ LO9. The student demonstrated an understanding of the practical material in the viva but not the theoretical aspects. The software has acceptable features. The software has acceptable performance and is reasonably stable but not necessarily foolproof or robust.
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40-49	LO1 – LO9. The project is complete, but nothing much above 'textbook'. Overall a fair piece of work but carried out to a minimal standard but sufficient to pass without further work. LO1. Fair work is shown in demonstrating the selection, justification and the application of appropriate methods, techniques and tools in tackling problems relating to the project. LO2. Fair work displayed in project management developing a project plan that scheduled the student's own activities and time relating to the project activities and timescale. LO3. Fair work demonstrated in the elicitation of the requirements of the project. A fair analysis and specification of the problem being solved demonstrated in some parts. Both the analysis and specification are not sufficiently well described, and even though it is possible to understand what has been solved in broad terms, a clear picture is missing. LO4. Background research presented basically. Some relevant background material is poorly presented (or is absent). Student is able to demonstrate a basic critical analysis broader issues and related literature to the project in part. Not very original. LO5. Technically fair and a but not too challenging a project. Minor development of existing skills. Minor development of new skills. Has needed significant help by supervisor. □ LO6. The student has demonstrated some understanding and wider consideration to legal, social, ethical, professional and sustainability issues that relate to the project. However, some of these may be underdeveloped. LO7. Main design decisions are described, but lacking in justification provided. Some testing of the software but lacking. LO8. weakly written in part, but still acceptable report. Some of the main themes and issues are not adequately identified. Poorly presented appraisal. The conclusions are brief and mostly non-informative evaluation includes few points but little critical analysis. Some quality work but insufficient in quantity. All sections are covered just competently. LO9. The student demonstrated an understanding of the practical material in the viva but not the theoretical aspects. The software has minimal to adequate features. The software is possibly weak but not entirely unacceptable, it tends to instability (not entirely) and lacks robustness in part.
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30-39	LO1 – LO9. The project is incomplete and the student has submitted work but cannot evaluate sufficiently. There is at least one section (specification, research, implementation, evaluation) that needs major reworking. A piece of work carried out to a standard that slightly underachieves the minimum standard expected but is recoverable. LO1. Insufficient work is shown in demonstrating the selection, justification and the application of appropriate methods, techniques and tools in tackling problems relating to the project. However, this is recoverable. LO2. Insufficient work displayed in project management developing a project plan that scheduled the student's own activities and time relating to the project activities and timescale. More time is required to recover some of the activities. LO3. Insufficient work demonstrated in the elicitation of the requirements of the project. Insufficient analysis and specification of the problem being solved demonstrated. However, all is recoverable. LO4. Background research presentation is very weak (or absent). Student is unable to demonstrate a satisfactory critical analysis broader issues and related literature to the project. Some of the analysis is presented in an incorrect or confusing manner but all is recoverable. LO5. Insufficiently technical or challenging a project as presented but recoverable. Weak in development of existing skills. Development of new skills weak. Has needed significant help by supervisor However, all this is recoverable. LO6. The student has not demonstrated sufficiently an understanding and wider consideration to legal, social, ethical, professional and sustainability issues that relate to the project but this is considered recoverable. LO7. Some of the main design decisions are weakly described. Testing of the software very incomplete (or minimal) with many missing test cases but recoverable. LO8. Weakly written, and partly incoherent report. Only a few main themes and issues are clearly identified but recoverable. Appraisal is either incorrectly presented or minimal and needs to improve. The conclusions are non-informative and some are wrong as currently stated. LO9. The student demonstrated a weak understanding of the work and was unable to explain sufficiently parts of it in the viva. The software has inadequate features. The performance is weak and the software is unstable.
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20-29	LO1 – LO9. A piece of work carried out to a standard that fails to achieve the required standard. The project is significantly incomplete. The project but made very weak effort for a major piece of work and is probably closer to a submitted coursework. LO1. Significant omissions in work shown in demonstrating the selection, justification and the application of appropriate methods, techniques and tools in tackling problems relating to the project. LO2. Poor work displayed in project management developing a project plan that scheduled the student's own activities and time relating to the project activities and timescale. Insufficient time to recover in remaining part of academic year. LO3. Significant omissions in work demonstrated in the elicitation of the requirements of the project. Significantly poor in analysis and specification of the problem being solved. LO4. Background research presentation is extremely weak (or absent). Student is unable to demonstrate sufficiently any critical analysis of broader issues and related literature to the project. LO5. Significantly low technically or non-challenging a project as presented. Hardly any development of existing skills. Development of new skills non-significant. Help from the supervisor mostly ignored. LO6. The student has failed to demonstrate any consideration to the wider consideration to legal, social, ethical, professional and sustainability issues that relate to the project. LO7. Most of the main design decisions are not described. Testing of the software too incomplete or not present. Appraisal is virtually non-existent. LO8. Very poorly written, and incoherent report. Significantly low main themes and issues identifiable. Very poorly written and incoherent report. The main themes and issues are not clearly identified. Most background material is not presented. Some of the presented background material contains factual errors. Most of the analysis is incorrect or confusing. The conclusions are non-informative and mostly wrong. LO9. The student demonstrated a poor understanding of the work and was unable to explain parts of it in the viva. The software has considerably fewer features than could be considered adequate. The software is highly unstable.
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1-19	LO1 – LO9. A piece of work carried out to a standard that fails to achieve the required standard significantly. The project is significantly incomplete. Misunderstood point of what the student had to do. The effort is closer to a bad coursework. □ LO1. Significantly poor work is shown in demonstrating the selection, justification and the application of appropriate methods, techniques and tools in tackling problems relating to the project. LO2. Significantly poor work displayed in project management developing a project plan that scheduled the student's own activities and time relating to the project activities and timescale. Insufficient time to recover in remaining part of academic year. LO3. Significant omissions in work demonstrated in the elicitation of the requirements of the project. Significantly poor in analysis and specification of the problem being solved. LO4. Background research presentation is of no real value to the project. Student is unable to demonstrate any critical analysis of broader issues and related literature to the project. LO5. Insufficiently technical or challenging a project. No development of existing skills. No Development of new skills. Help from the supervisor ignored or not sought. LO6. The student has significantly failed to mention any of the wider consideration to legal, social, ethical, professional and sustainability issues that relate to the project. LO7. The main design decisions are not described. Testing of the software too not present. Appraisal is non-existent. LO8. Very poorly written and incoherent report. The main themes and issues are not identified at all. Most background material is not presented. Most of the presented background material contains factual errors. The analysis is incorrect or confusing. Main design decisions are not described. Testing of the software too incomplete or not present. No appraisal. No conclusions, or the conclusions are wrong. LO9. The student demonstrated a poor understanding of the work and was unable to explain parts of it in the viva. The software does not do almost anything useful and has almost no appropriate features, if any.
1	In cases of proven plagiarism to indicate attempt
0	Only in cases of non-submission

How will this module enhance employability?

Students will be encouraged to explore the job market for potential jobs in their areas of interest. The student will propose a project that may be added to a portfolio they can demonstrate to a potential employer. The Final Project can be used by the student to convince a potential employer that they have the required skills and domain knowledge to undertake the work for a targeted job.